InTraScale: Accelerating Video Generation via In-Trajectory Resolution Scaling

Anonymous submission

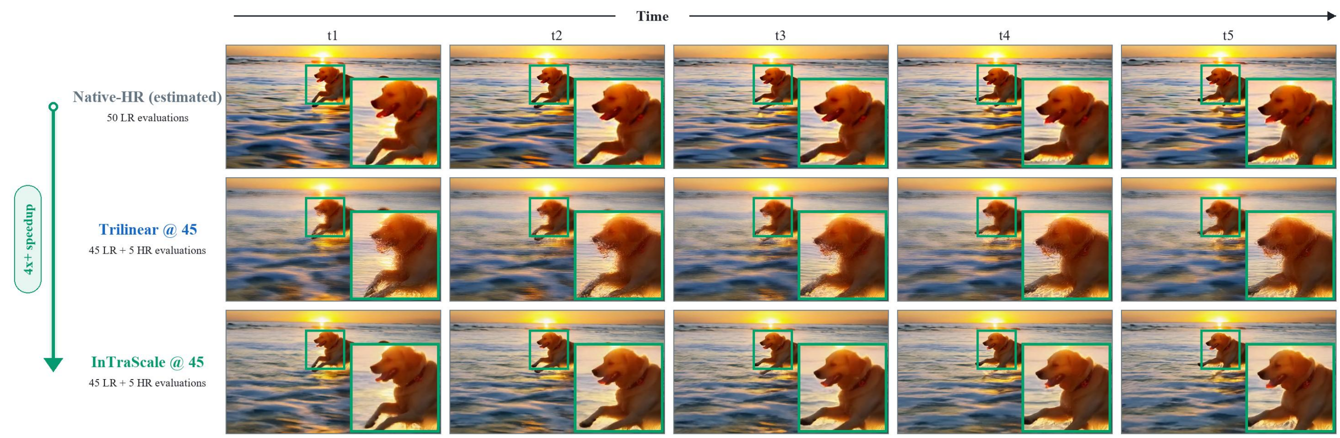

Figure 1: **Real-video temporal comparison of the resolution transition.** Five matched frames compare a content-aligned Native-HR (estimated) structural reference, trilinear interpolation at evaluation 45, and InTraScale at the same transition point.

Abstract

High-resolution video diffusion remains expensive because every denoising evaluation processes the full target-resolution spatiotemporal latent sequence. In-trajectory resolution scaling reduces this cost by executing an inexpensive low-resolution prefix and reserving target-resolution computation for the remaining evaluations, without launching a separate super-resolution diffusion process or increasing the total denoising-evaluation budget. However, fixed trilinear interpolation cannot recover the VAE-induced cross-resolution latent correspondence, while a learned upsampler faces a training–inference mismatch between completed clean latents and predictions from an unfinished trajectory. We introduce InTraScale, a decoupled framework that addresses these challenges while keeping the pretrained generator frozen. Its In-Trajectory Upsampler learns the cross-resolution correspondence from content-aligned clean latent pairs, whereas its Trajectory-Tail Distillation mechanism transfers the corrective effect of the omitted low-resolution trajectory tail to the transition evaluation, producing a more endpoint-like input. A scheduler-consistent state reconstruction then returns the prediction to the remaining target-resolution trajectory. On a 50-evaluation Wan2.1 sampler, InTraScale achieves $3.21\times$ and $4.44\times$ warm-model speedups while retaining 97.76% and 97.53% of the Native-HR VBench-5 score, respectively. On a distilled four-evaluation model, it provides an additional $2.13\times$ acceleration while retaining 99.58% of the native distilled score. These results establish in-trajectory resolution scaling as an effective spatial acceleration strategy complementary to timestep distillation.

1 Introduction

Recent video diffusion models generate increasingly coherent motion and fine-grained visual content, but their computational cost remains a major obstacle to practical deployment (Wan et al. 2025). Each denoising evaluation applies a large diffusion transformer to the complete spatiotemporal latent sequence, making target-resolution generation particularly expensive. Timestep distillation reduces the number of evaluations, but every retained evaluation still processes the full target-resolution grid. Spatial acceleration is therefore complementary to temporal acceleration: global layout, semantic structure, and motion can be established economically at low resolution, while expensive target-resolution computation is reserved for recovering local structure and high-frequency details.

Such low-to-high-resolution pipelines can follow two schedules, depending on whether resolution is increased after low-resolution generation or before its trajectory is complete. Most existing pipelines follow the former, completing a low-resolution trajectory before applying an RGB-space or latent-space upsampling stage (Ho et al. 2022; Blattmann et al. 2023; Wang et al. 2023b). RGB-space refinement may incur codec and state-reconstruction overhead when its output is returned to a generative model, whereas latent cascades avoid this conversion by directly upsampling the completed representation (Xie et al. 2025; Zhao et al. 2026). Both nevertheless append refinement only after low-resolution generation has completed. In-trajectory resolution scaling follows the latter: it transitions earlier and uses the remaining denoising evaluations as the high-resolution continuation (Jeong et al. 2025; ModelTC 2026), avoiding a separate super-resolution diffusion process or an increase in the original evaluation budget.

Moving the resolution transition inside the denoising trajectory, however, creates two coupled representation challenges. Simple latent interpolation only resamples the spatial grid and cannot recover the cross-resolution correspondence induced by the pretrained VAE. The resulting structural distortion becomes particularly difficult to correct under a late transition, where only a short high-resolution continuation remains. A natural alternative is to replace interpolation with a learned latent upsampler. However, the upsampler is typically trained using completed, content-aligned clean latent pairs, whereas in-trajectory inference provides a clean prediction derived from an unfinished low-resolution trajectory. This prediction retains a timestep-dependent residual that the omitted low-resolution trajectory tail would otherwise remove. Directly applying the learned upsampler therefore forces it to perform both cross-resolution reconstruction and unfinished trajectory correction, creating a training–inference mismatch. These two functions should consequently be modeled separately.

To address these problems, we introduce **InTraScale**, a decoupled framework for learned in-trajectory resolution scaling. The **In-Trajectory Upsampler (ITU)** learns the VAE-specific correspondence between completed clean low- and high-resolution latent pairs, replacing geometry-only interpolation with learned cross-resolution reconstruction. **Trajectory-Tail Distillation (TTD)** separately transfers the corrective effect of the omitted low-resolution trajectory tail to the transition evaluation, producing a more endpoint-like clean prediction for ITU. InTraScale then reconstructs the next target-resolution scheduler state and executes the remaining evaluations using the frozen flow predictor. On conventional 50-evaluation Wan2.1 sampling, InTraScale achieves $3.21\times$ and $4.44\times$ warm-model speedups while retaining 97.76% and 97.53% of the Native-HR VBench-5 score, respectively. Applied to a distilled four-evaluation model, it provides an additional $2.13\times$ acceleration while retaining 99.58% of the native distilled-model score.

Our contributions are summarized as follows:

- We introduce InTraScale, a learned in-trajectory resolution-scaling framework that distributes a fixed denoising-evaluation budget between an inexpensive low-resolution prefix and a high-resolution continuation.
- We decouple resolution conversion from unfinished trajectory correction: ITU learns the VAE-specific cross-resolution clean-latent correspondence, while TTD distills the omitted low-resolution trajectory tail into the transition evaluation.
- We evaluate InTraScale on conventional 50-evaluation and distilled four-evaluation Wan2.1 samplers, demonstrating substantial spatial acceleration both before and after temporal distillation.

2 Related Work

Post-Generation Resolution Enhancement. Post-generation methods enhance a completed low-resolution output in RGB space (Zhou et al. 2024; Xu et al. 2024; He et al. 2024) or latent space (Xie et al. 2025; Zhao et al. 2026). MrFlow combines pixel-space super-resolution with subsequent high-resolution refinement for image flow-matching models (Zheng et al. 2026b), while FlashVSR distills diffusion-based restoration into a one-step streaming model (Zhuang et al. 2026). Despite different representations and inference costs, these methods operate after the low-resolution trajectory has completed, whereas InTraScale reduces target-resolution computation within the original generative trajectory.

Multiscale and Resolution-Changing Generation. Matryoshka Diffusion and Pyramidal Flow Matching train generators around multiscale denoising processes (Gu et al. 2024; Jin et al. 2025). For pretrained models, RALU and LightX2V prescribe variable-resolution inference schedules (Jeong et al. 2025; ModelTC 2026), while Fresco and Spectral Progressive Diffusion progressively expand noisy states using spatial or spectral constructions (Zheng et al. 2026a; Xiao et al. 2026). These approaches either incorporate multiscale generation during training or define how noisy latent states are expanded. InTraScale instead keeps the generator frozen, learns the VAE-specific correspondence between clean low- and high-resolution latents, and separately corrects the unfinished low-resolution trajectory tail.

Orthogonal Video Diffusion Acceleration. Dynamic-computation methods vary model budgets or patch sizes across denoising evaluations and spatial regions (Anagnostidis et al. 2025; Kim, Ghadiyaram, and Gadde 2026; Li et al. 2026). Temporal acceleration reduces the number of evaluations through consistency or transition distillation (Wang et al. 2023a; Li et al. 2024; Ding et al. 2025; Nie et al. 2026), whereas caching methods reuse attention maps or intermediate features (Zhao et al. 2025; Liu et al. 2025). DisCa further learns caching under a restricted distilled trajectory (Zou et al. 2026). InTraScale instead reduces the spatial cost of each retained evaluation and is therefore complementary to temporal distillation and feature reuse, as demonstrated on a distilled four-evaluation generator.

3 Method

This section develops InTraScale by decomposing the transition into cross-resolution reconstruction and unfinished-trajectory correction. Accordingly, ITU learns the clean-

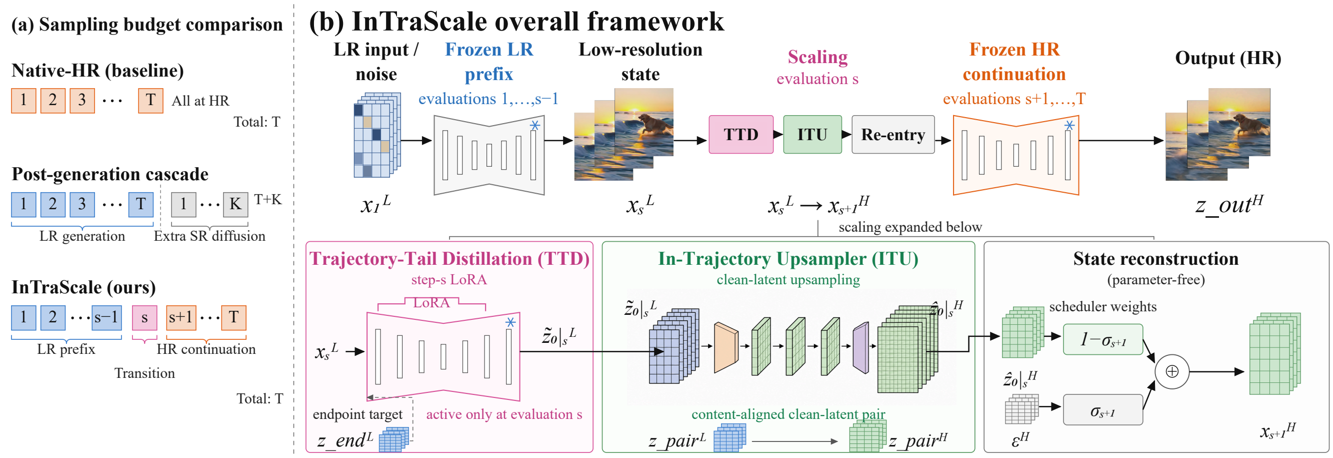

Figure 2: **Sampling budget and overall framework of InTraScale.** (a) Native-HR executes all T evaluations at target resolution, whereas a post-generation cascade appends K super-resolution evaluations after completing low-resolution generation. (b) At evaluation s, TTD distills the omitted low-resolution trajectory tail into an endpoint-aligned clean prediction. ITU, trained on content-aligned clean-latent pairs, lifts this prediction to the target-resolution latent space.

latent correspondence across resolutions, TTD produces a more endpoint-like transition prediction, and parameter-free state reconstruction returns the lifted latent to the target-resolution trajectory.

3.1 Challenges of In-Trajectory Transition

Latent distortion from trilinear interpolation. A fixed trilinear interpolation operator only resamples the spatial grid; it does not recover the correspondence between the low- and high-resolution representations induced by the pre-trained VAE. Consequently, interpolated latents may exhibit distorted structures and inaccurate high-frequency statistics on the target-resolution grid. RALU reported that late latent upsampling introduces aliasing artifacts, particularly around high-frequency edge regions. This exposes a fundamental quality–efficiency trade-off: a later resolution transition provides greater acceleration but leaves fewer high-resolution evaluations to correct the errors introduced during resolution conversion. With only a short high-resolution suffix, the remaining target-resolution evaluations may therefore be insufficient to recover structural details lost or distorted by trilinear interpolation.

Training–inference mismatch for learned upsampling. Replacing interpolation with a learned latent upsampler introduces an input distribution shift. A latent upsampler is naturally trained using paired clean latents obtained from completed, content-aligned low- and high-resolution videos. Its low-resolution training input is thus an endpoint-like clean representation containing all spatial and temporal details supported by the low-resolution bandwidth. During in-trajectory inference, however, the upsampler does not receive such a converged latent. Here, x_s^L denotes the low-resolution latent state at step s, v_ϕ is the frozen flow predictor with parameters ϕ, c is the text condition, and σ_s is the scheduler coefficient. We use the subscript $0|s$ to denote a clean latent predicted from the state at step s. For a resolution transition after evaluation s, the upsampler receives a single-evaluation clean prediction derived from an unfinished low-resolution trajectory:

$$\hat{z}_{0|s}^{L} = x_s^L - \sigma_s v_\phi(x_s^L, s, c).$$

It contains a timestep- and scheduler-dependent residual that would otherwise be removed by the remaining low-resolution evaluations. Its distribution therefore differs from that of the clean low-resolution latents used to train the upsampler. Directly applying the upsampler to this prediction forces a single module to perform both cross-resolution reconstruction and unfinished trajectory correction, weakening its ability to preserve structure across scales.

These two challenges motivate the decoupled design of InTraScale. The In-Trajectory Upsampler learns the cross-resolution clean-latent correspondence, while Trajectory-Tail Distillation transfers the corrective effect of the omitted low-resolution trajectory tail to the final low-resolution evaluation, producing a more endpoint-like input for the subsequent resolution transition.

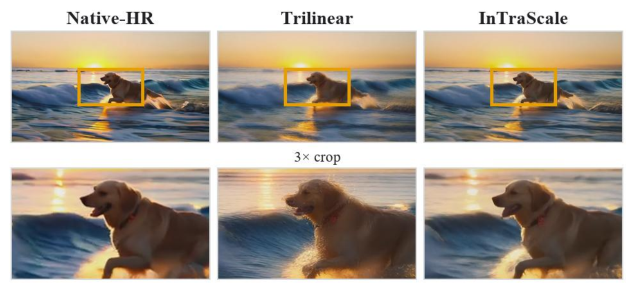

Figure 3: Trilinear interpolation under a late resolution transition. The columns show content-aligned Native-HR, trilinear interpolation, and InTraScale outputs at evaluation 45.

Figure 4: Unfinished-trajectory residual at evaluation 45, the final low-resolution evaluation.

3.2 Overall Framework

Figure 2 summarizes the sampling budget and the decoupled pipeline motivated by the two challenges above. We now formalize its components and trajectory states. Let $v_\phi(x_k^R, k, c)$ denote the frozen flow predictor, where $R \in \{L, H\}$ indicates the low- or target-resolution latent space, x_k^R is the latent state presented to the k-th denoising evaluation, and c is the text condition. We use σ_k for the corresponding scheduler coefficient and T for the total number of denoising evaluations. We refer to the pretrained flow predictor and VAE collectively as the frozen target model. We define a resolution transition after evaluation s. The s-th evaluation is therefore the final evaluation executed on the low-resolution grid. Evaluations $1, \dots, s-1$ form the frozen low-resolution prefix, the TTD update is applied only during evaluation s, and evaluations $s+1, \dots, T$ form the frozen high-resolution continuation.

During the final low-resolution evaluation, the prediction $\hat{z}_{0|s}^L$ introduced in the preceding subsection is not directly passed to the upsampler. Trajectory-Tail Distillation (TTD) first produces an endpoint-aligned clean prediction $\tilde{z}_{0|s}^L$. The In-Trajectory Upsampler (ITU), parameterized by ψ, then maps this prediction to the target-resolution clean latent $\tilde{z}_{0|s}^H = U_\psi(\tilde{z}_{0|s}^L)$.

To continue the original denoising trajectory, InTraScale reconstructs the next target-resolution scheduler state as

$$x_{s+1}^H = (1 - \sigma_{s+1})\tilde{z}_{0|s}^H + \sigma_{s+1}\epsilon^H, \qquad \epsilon^H \sim \mathcal{N}(0, I),$$

where ϵ^H is target-resolution noise determined by the generation seed. This parameter-free operation matches the next scheduler noise level without auxiliary diffusion or a decode–encode round trip. We require $s < T$, so the frozen predictor performs at least one target-resolution evaluation. This high-resolution tail completes structures and textures underdetermined by the low-resolution latent. The complete inference pipeline is

$$x_1^L \xrightarrow{\text{LR}} x_s^L \xrightarrow{\text{TTD}} \tilde{z}_{0|s}^L \xrightarrow{\text{ITU}} \tilde{z}_{0|s}^H \xrightarrow{\text{reconstruction}} x_{s+1}^H \xrightarrow{\text{HR}} z_{\text{out}}^H.$$

Here, z_{out}^H denotes the final target-resolution clean latent, which is decoded into the output video by the frozen VAE.

The ITU and TTD parameters are trained independently using supervision generated by the frozen target model. ITU learns the cross-resolution clean-latent correspondence, whereas the TTD update learns to remove the unfinished low-resolution trajectory residual at the transition point.

3.3 In-Trajectory Upsampler

The In-Trajectory Upsampler (ITU) addresses the cross-resolution representation mismatch identified in Section 3.1. Rather than treating latent upsampling as spatial interpolation, ITU learns the correspondence between clean low- and high-resolution representations induced by the frozen target model.

Generator-endogenous supervision. We construct the training pairs using videos generated by the frozen target model. Given a generated target-resolution video v^H, we first obtain its content-aligned low-resolution counterpart through RGB-space downsampling:

$$v^L = \mathcal{D}_{\text{rgb}}(v^H).$$

The two videos are then encoded independently by the same frozen VAE $\mathcal{E}$:

$$z_{\text{pair}}^L = \mathcal{E}(v^L), \qquad z_{\text{pair}}^H = \mathcal{E}(v^H).$$

The resulting pair $(z_{\text{pair}}^L, z_{\text{pair}}^H)$ shares the same semantics, frame count, and motion trajectory. Performing the resolution change before VAE encoding allows ITU to learn the target model's actual cross-resolution latent correspondence instead of imitating a predefined interpolation operator in latent space.

Spatiotemporal latent upsampling. ITU maps a clean low-resolution latent to the target-resolution latent grid while preserving the temporal dimension:

$$U_\psi : \mathbb{R}^{B \times 16 \times F \times h_L \times w_L} \to \mathbb{R}^{B \times 16 \times F \times h_H \times w_H},$$

where B and F denote the batch size and number of latent frames, respectively. The network follows an encoder–resampler–decoder design. A 3D convolution first projects the input into a 256-channel hidden representation, followed by four spatiotemporal residual blocks. A learned spatial lifting operator then changes only the height and width. Four additional residual blocks and a final 3D convolution reconstruct the 16-channel target latent. The 3D blocks allow adjacent latent frames to contribute to the reconstruction without changing the temporal resolution.

We instantiate the spatial lifting operator according to the required grid ratio. For $480 \times 832 \to 720 \times 1248$, the latent grid changes from 60×104 to 90×156. We realize this $3/2$ ratio using a 3× spatial PixelShuffle followed by anti-aliased stride-2 downsampling. For $368 \times 640 \to 720 \times 1248$, a 2× PixelShuffle produces a 92×160 grid, which is center-cropped to 90×156. The resolution-specific lifting operators determine the target geometry, while the surrounding spatiotemporal network learns the latent correspondence.

Cross-resolution objective. Given

$$\hat{z}_{\text{pair}}^H = U_\psi(z_{\text{pair}}^L),$$

We train ITU with

$$\mathcal{L}_{\text{ITU}} = \mathcal{L}_{\text{rec}} + 0.2\mathcal{L}_{\text{lf}} + 0.1\mathcal{L}_{\text{temp}}.$$

Here, $\mathcal{L}_{\text{rec}}$ is the Charbonnier reconstruction loss, $\mathcal{L}_{\text{lf}}$ enforces consistency between the spatially downsampled prediction and the low-resolution input, and $\mathcal{L}_{\text{temp}}$ matches first-order temporal differences between adjacent latent frames. We use a Charbonnier constant of $\epsilon = 10^{-3}$. Here, $\mathcal{D}_{\text{lat}}$ maps the prediction back to the low-resolution latent grid, and Δ_t denotes first-order differences between adjacent latent frames. We use $\lambda_{\text{lf}} = 0.2$, $\lambda_{\text{temp}} = 0.1$, and $\epsilon = 10^{-3}$. The three terms constrain target-latent reconstruction, low-frequency content preservation, and temporal evolution, respectively.

A low-resolution latent does not uniquely determine every high-frequency detail in its target-resolution counterpart. ITU therefore focuses on transferring structure, motion, and latent statistics to the target grid. The remaining frozen high-resolution trajectory retains the generative responsibility for details that are underdetermined by the low-resolution input.

3.4 Trajectory-Tail Distillation

ITU is trained on completed clean-latent pairs, whereas its inference input is predicted from an unfinished low-resolution trajectory. TTD reduces this deployment mismatch by distilling the omitted low-resolution trajectory tail into the final low-resolution evaluation. The aligned prediction is subsequently lifted by ITU and returned to the native high-resolution schedule.

For endpoint supervision, we run the frozen target model for each training sample using a fixed prompt, generation seed, initial noise, scheduler, and guidance configuration. We cache the state x_s^L immediately before the final low-resolution evaluation and continue the same low-resolution trajectory to completion. We denote the resulting clean endpoint by z_{end}^L. The pair

$$(x_s^L, z_{\text{end}}^L)$$

therefore preserves the semantic and stochastic conditions at the transition point while providing direct supervision for the trajectory residual remaining at evaluation s.

We implement TTD through a step-specific low-rank update $\Delta\theta_s$ to the frozen flow predictor. For a linear layer W, the LoRA parameterization is

$$W_\lambda = W + \lambda \frac{\alpha}{r} BA,$$

where r is the LoRA rank, α is its training scale, and λ controls the TTD update strength during inference. We use rank $r = 32$ and inject the update into the q, k, v, and o attention projections and the two feed-forward projections. The base parameters ϕ remain frozen.

At evaluation s, the TTD-augmented predictor produces

$$\tilde{z}_{0|s}^L = x_s^L - \sigma_s v_{\phi+\lambda\Delta\theta_s}(x_s^L, s, c).$$

We train the TTD parameters to regress the corresponding native low-resolution endpoint using

$$\mathcal{L}_{\text{TTD}} = \mathcal{L}_1 + 0.1\mathcal{L}_{\text{MSE}} + \lambda_{\text{tt}}\mathcal{L}_{\text{temp}}.$$

Here, $\mathcal{L}_1$ and $\mathcal{L}_{\text{MSE}}$ compare $\tilde{z}_{0|s}^L$ with z_{end}^L, while $\mathcal{L}_{\text{temp}}$ matches their first-order temporal differences. We set $\lambda_{\text{tt}} = 0.05$ for the earlier transition after evaluation $s = 40$ in the 50-step sampler and zero for the later and distilled settings. We select the inference strength $\lambda = 0.75$ on held-out development prompts and keep it fixed for all reported samples.

The TTD update is applied only during evaluation s. Its strength is zero for the preceding evaluations, so the frozen low-resolution prefix and the cached pre-transition state x_s^L remain unchanged under matched deterministic conditions. TTD is therefore a transition-localized distillation update rather than a general low-resolution step-reduction model.

4 Experiments

We evaluate whether InTraScale improves the quality–efficiency trade-off under conventional 50-evaluation sampling, remains effective after four-step timestep distillation, and addresses the transition failures identified in Section 3.1. We report system-level comparisons followed by component and design analyses.

4.1 Experimental Setup

We evaluate InTraScale on both the standard 50-evaluation Wan2.1 T2V-1.3B sampler and a four-evaluation Wan2.1 T2V-14B StepDistill-CfgDistill sampler. This section first examines the 50-evaluation setting, where resolution transitions are placed after evaluation 40 or 45. We additionally evaluate trilinear interpolation after evaluations 40, 45, and 48 to characterize the effect of increasingly delayed transitions. All configurations generate 81-frame videos at 720×1248 resolution from matched text prompts, random seeds, initial noise, schedulers, and guidance settings.

We evaluate generation quality on ten matched prompts using VBench. Our primary aggregate metric, VBench-5, is the unweighted mean of subject consistency, background consistency, motion smoothness, aesthetic quality, and imaging quality. Dynamic degree is reported separately because a larger motion magnitude does not necessarily indicate better quality. Prompt-level differences are analyzed using paired bootstrap confidence intervals and two-sided exact sign tests.

We measure efficiency using single-video warm-model end-to-end latency. Each configuration is loaded once, followed by one warm-up video and five measured videos in the same resident process. Measurements are CUDA-synchronized and include all per-video denoising evaluations, the TTD update and ITU execution, scheduler-state reconstruction, VAE processing, and output operations. One-time runner construction and checkpoint loading are excluded.

4.2 Overall Quality–Efficiency Trade-off

We compare InTraScale with three classes of alternatives. Native-HR executes all 50 evaluations at the target resolution. Trilinear directly resamples the latent grid at different transition points. RALU-style rescheduling adjusts the noise–timestep correspondence at the representative transition after evaluation 45. Finally, Endpoint-ITU completes all 50 evaluations at low resolution, applies ITU to the resulting endpoint, and performs five additional high-resolution refinement evaluations. Figure 5 visualizes the resulting operating points, while Table 1 reports their exact values.

Method	LR/HR evaluations	Warm latency (s) ↓	Speedup ↑	VBench-5 ↑
Native-HR	0/50	187.54 ± 0.26	$1.00\times$	**0.82836**
Trilinear@40	40/10	58.12 ± 0.20	$3.23\times$	0.77812
InTraScale@40	40/10	58.49 ± 0.16	$\mathbf{3.21}\times$	**0.80983**
Trilinear@45	45/5	41.98 ± 0.19	$4.47\times$	0.76776
RALU-style@45	45/5	42.01 ± 0.19	$4.46\times$	0.80440
InTraScale@45	45/5	42.26 ± 0.20	$\mathbf{4.44}\times$	**0.80792**
Trilinear@48	48/2	32.27 ± 0.18	$5.81\times$	0.76409
Endpoint-ITU, 5 HR	50/5	44.00 ± 0.27	$4.26\times$	0.80073

Table 1: Warm-model quality–efficiency comparison for the 50-evaluation sampler.

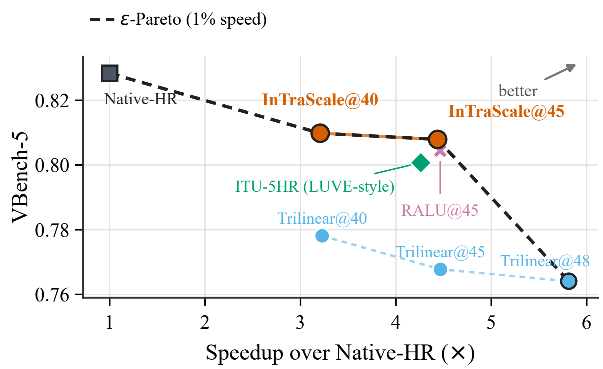

Figure 5: **Wan50 quality–efficiency Pareto frontier.**

InTraScale@40 reduces warm-model latency from 187.54 to 58.49 seconds, corresponding to a $3.21\times$ speedup, while retaining 97.76% of the Native-HR VBench-5 score. Delaying the transition to evaluation 45 further reduces latency to 42.26 seconds, achieving a $4.44\times$ speedup and retaining 97.53% of the Native-HR score. Moving the transition from evaluation 40 to 45 reduces latency by an additional 27.74%, with a VBench-5 decrease of only 0.00191. The two operating points therefore provide a controllable quality–efficiency trade-off.

At matched transition positions, the learned transition introduces little additional latency over trilinear interpolation. InTraScale@40 and InTraScale@45 add only 0.63% and 0.67% latency, respectively, while improving VBench-5 by 0.03171 and 0.04016. The widening quality gap at the later transition supports the challenge identified in Section 3.1: as the high-resolution continuation becomes shorter, trilinear interpolation leaves insufficient evaluations to repair the induced latent distortion.

At evaluation 45, InTraScale also improves VBench-5 over RALU-style rescheduling by 0.00352 with a 0.61% latency increase. This comparison suggests that scheduler adaptation alone does not fully recover the cross-resolution latent correspondence addressed by ITU.

The Endpoint-ITU baseline completes the entire low-resolution trajectory before resolution conversion and adds five high-resolution refinements. Despite executing these additional evaluations, it reaches only 0.80073 VBench-5 at 44.00 seconds. InTraScale@45 is 3.96% faster and improves VBench-5 by 0.00719. Transitioning before the low-resolution endpoint therefore provides a more favorable operating point than completing low-resolution generation and subsequently launching a separate high-resolution refinement suffix.

Overall, InTraScale@40 and InTraScale@45 establish two non-dominated operating points between Native-HR quality and aggressive low-resolution execution. The prompt-paired confidence intervals against Native-HR overlap zero, but the evaluation does not constitute an equivalence test. We therefore characterize InTraScale as a quality–efficiency frontier rather than claiming lossless acceleration.

4.3 Further Accelerating Distilled Video Generation

Although Section 4.2 demonstrates substantial acceleration for conventional 50-evaluation sampling, deployment-oriented video generators increasingly rely on timestep-distilled models to reduce the trajectory length. We therefore ask whether in-trajectory resolution scaling can provide additional gains after temporal computation has already been compressed. To this end, we apply InTraScale to a four-evaluation Wan2.1 T2V-14B StepDistill-CfgDistill model. The resolution transition occurs after the third low-resolution evaluation, leaving only one target-resolution evaluation for generative refinement.

We compare against Native-HR4, which executes all four evaluations at the target resolution, and Trilinear@3, which uses the same 3-LR/1-HR schedule as InTraScale but replaces ITU with fixed trilinear interpolation and disables TTD. We further construct two controlled post-generation baselines on the same distilled backbone. Both complete all four low-resolution evaluations before increasing the spatial resolution. LUVE-style then applies learned latent upsampling followed by two high-resolution refinements, whereas MrFlow-style uses a decode–RGB-upsample–re-encode path followed by one high-resolution correction. These configurations reproduce the corresponding computational patterns rather than the complete original systems. Figure 6 visualizes these operating points, while Table 2 reports their exact values.

InTraScale-D4 reduces warm-model latency from 42.49 to 19.92 seconds, providing an additional $2.13\times$ acceleration over Native-HR4. Meanwhile, its VBench-5 decreases by only 0.00361, corresponding to a relative reduction of 0.42% and a score retention of 99.58%. The prompt-paired

Method	LR/HR evaluations	VBench-5 ↑	Temporal ↑	Warm latency (s) ↓	Speedup
Native-HR4	0/4	**0.86041**	**0.98412**	42.49	1.00×
Trilinear@3	3/1	0.81796	0.97400	**18.33**	**2.32**×
InTraScale-D4@3	3/1	0.85680	**0.97663**	19.92	**2.13**×
LUVE-style	4/2	0.85981	0.97331	28.76	1.48×
MrFlow-style	4/1	0.85903	0.97484	27.12	1.57×

Table 2: Quality and warm-model efficiency under aggressive four-evaluation sampling.

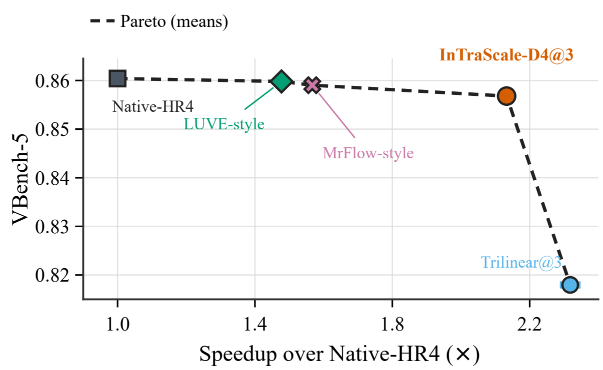

Figure 6: **Distill4 quality–efficiency Pareto frontier.**

confidence interval overlaps zero; we therefore describe this difference as marginal rather than claiming quality equivalence.

Trilinear@3 reaches a slightly higher $2.32\times$ speedup, but its VBench-5 drops by 0.04244 relative to Native-HR4. At the same 3-LR/1-HR schedule, InTraScale adds only 1.59 seconds over Trilinear@3 while improving VBench-5 by 0.03884 and temporal consistency by 0.00264. This result strengthens the observation from the 50-evaluation experiments: when the high-resolution continuation is reduced to a single evaluation, trilinear interpolation leaves insufficient correction capacity, whereas the learned transition preserves most of the native quality.

InTraScale also provides a more favorable operating point than the post-generation alternatives. It reduces latency by 30.73% relative to LUVE-style and by 26.53% relative to MrFlow-style, while its VBench-5 is lower by only 0.00301 and 0.00223, respectively. The quality difference from MrFlow-style has a prompt-paired confidence interval that overlaps zero; the LUVE-style advantage is small but detectable. InTraScale additionally obtains the highest temporal-consistency score among the accelerated alternatives shown in Table 2.

These results demonstrate that in-trajectory resolution scaling is complementary to timestep distillation. Even after the denoising trajectory has been compressed to four evaluations, InTraScale further reduces spatial computation by more than half while retaining nearly all of the native distilled-model quality.

4.4 Component and Design Analysis

We next examine whether ITU and TTD address the two transition-specific challenges identified in Section 3.1. On 50 completed clean-latent pairs per scale, ITU consistently improves cross-resolution reconstruction over trilinear interpolation. For 480p $\rightarrow$ 720p, it reduces latent L_1, LPIPS, and temporal L_1 by 48.6%, 73.3%, and 30.8%, respectively. Under the more aggressive 368p $\rightarrow$ 720p conversion, the corresponding reductions are 33.4%, 51.1%, and 15.5%. These results confirm that ITU learns the VAE-specific cross-resolution correspondence rather than merely approximating geometric interpolation.

TTD consistently brings the transition-point prediction closer to the endpoint of the complete low-resolution trajectory. It reduces endpoint latent L_1 by 25.82%, 21.03%, and 5.03% at Wan50@40, Wan50@45, and Distill4@3, respectively, with improvements on all ten samples in each setting. The factorial evaluation further separates the roles of the two modules: replacing trilinear interpolation with ITU improves VBench-5 by 0.03085–0.04066, whereas enabling TTD changes the aggregate score by only -0.00050 to $+0.00087$. ITU therefore provides the dominant perceptual gain, while TTD primarily corrects the transition-specific endpoint mismatch.

Finally, we validate the target-resolution state reconstruction on an independent eight-prompt set. Using seed-derived target-resolution noise instead of resized low-resolution flow improves VBench-5 by 0.02483 and temporal consistency by 0.01605, showing the importance of reconstructing a scheduler-consistent high-resolution state. Performance remains stable when the TTD update strength varies from 0.25 to 1.0; we select $\lambda = 0.75$ on the validation set and keep it fixed for all test evaluations. Full metric breakdowns and strength sweeps are provided in the supplementary material.

5 Conclusion

We presented InTraScale, which accelerates high-resolution video generation by scaling resolution within a fixed denoising budget, without a separate super-resolution diffusion process. TTD adapts the transition state, while ITU performs VAE-specific cross-resolution latent lifting, allowing the frozen generator to continue at the target resolution. On 50-evaluation Wan2.1 sampling, InTraScale achieves $3.21\times$ and $4.44\times$ warm-model speedups while retaining 97.76% and 97.53% of the Native-HR VBench-5 score. On a four-evaluation distilled model, it achieves $2.13\times$ acceleration with 99.58% score retention. Ablations confirm the complementary roles of TTD and ITU, establishing in-trajectory resolution scaling as a spatial acceleration strategy complementary to timestep distillation.

Limitations. Model-endogenous self-distillation specializes ITU to the VAE and resolution pair and TTD to the transition evaluation. This enables close adaptation without external paired data, but new configurations may require renewed supervision and module adaptation. Future work will explore cross-configuration transfer and content-aware transition policies.

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